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EXPERIMENTAL INVESTIGATION OF THE CHEMISTRY OF EXCITED STATES OF RARE GASES

MASTER

Fourth Quarterly (and Annual) Progress Report
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Cumulative Listing of Published Papers

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5. H. C. Brashears, Jr., D. W. Setser and D. DesMarteau, "Vacuum-ultraviolet Photolysis of XeF_2: XeF(B) Fluorescence and Quenching," Chem. Phys. Lett., 48, 84 (1977).

6. J. E. Velazco, "Quenching Reactions of $Xe(^3P_2)$ Reactions," Ph. D. Dissertation, Kansas State University (1977).

7. J. H. Kolts, H. C. Brashears and D. W. Setser, "Redetermination of the $N_2(C)$-$N_2(B)$ Branching Ratio from the $Ar(^3P_{0,2})$ Reaction, J. Chem. Phys. 67, 2931 (1977).

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11. R. S. F. Chang and D. W. Setser, "Lifetime and Two-body Deactivation Rate Constants for Ar(4p) States", J. Chem. Phys. 69, 3885 (1978).

12. H. C. Brashears, Jr. and D. W. Setser, "Transfer and Quenching Rate Constants for XeF(III,1/2) and XeF(II,3/2)." App. Phys. Lett. 33, 821 (1978).

13. D. W. Setser, T. D. Dreiling and J. H. Kolts, "Product Distributions from Quenching of Metastable Rare Gas Atoms,"J. Photochem. 9, 91 (1978). Presented at IX International Conference on Photochemistry, Cambridge, England.

Summary

During this quarterly period, Mr. Harold Brashears (graduate student) worked full time. Mr. Brashears completed his study of the quenching of XeF(B) using the steady-state photolysis of XeF_2 as the source of XeF(B). Harold has been successful in observing the XeF(C) state, as well as the XeF(B) state, by using a very clean CO(A-X) resonance lamp to photolyze the XeF_2. We have found that XeF(C) must be included in the kinetic scheme in order to understand the quenching of XeF*. A reprint of Harold's work is attached to this report. In addition to the published work, Harold has measured the dependence of the sum of I(B-X) and I(C-A) upon added reagent bath gas pressure. By combining all of the results we can obtain quenching rate constants (as well as transfer rate constants) of XeF(B) and XeF(C) for favorable cases. Mr. Brashears is developing techniques for a photolysis of KrF_2 so that quenching and transfer of KrF(B) can be studied. Harold completed the photodissociation study of XeO_4 and the investigation of the reactions of rare gas metastable atoms with XeO_4 (see attached report). Unfortunately none of the excitation methods gave the rare gas oxide (Rg^+O^-) emission that we were seeking.

Dr. Bob Chang continued using the pulsed laser induced fluorescence technique to measure radiative lifetimes and two-body quenching rate constants of rare gas excited states from July 15-Aug. 15. On August 15 he moved to the Naval Labs (Washington), where he is working on excimer laser development. Bob's study of Ne(3p)** in Ne is complete and in manuscript form. Dr. Horiguchi has been hired as a replacement postdoctoral fellow. Dr. Horiguchi did some excellent studies of E-V transfer in $Hg(^3P_0)$-CO and NO systems in Tokyo using the phase shift technique. Dr. Horiguchi has been learning the pulsed techniques so that we can continue our laser induced fluorescence measurements. We hope to acquire additional data for Kr(5p)** in Ar and Xe(6p)** in Ar before publishing those results. Obtaining these results is Dr. Horiguchi's first task.

The principal investigator gave an invited lecture to the International Photochemistry Conference in mid-August. Also considerable time has been spent on a review chapter concerned with generation and detection of excited states in flow systems; it is nearly completed. The three papers which were published this quarter and the report on XeO_4 are attached as reprints to complete this final report.

Studies of Vacuum U-V Photolysis and Reactions of Rare Gas Metastable Atoms with XeO_4

H. C. Brashears

Introduction

This is a report of our investigation of the reactions of $Xe(^3P_2)$ and $Ar(^3P_2)$ with XeO_4 and the vacuum U.V. photolysis of XeO_4. The objective was to ascertain whether or not electronically excited XeO* was produced by the photolytic decomposition of XeO_4 or by reaction with metastable Ar or Xe. By analogy to XeF_2 and KrF_2, both of which give high yields of XeF(B) and KrF(B) upon either photolysis or reaction with metastable states of rare gases, it was hoped that the first excited coulombic state of XeO** would be produced. The bound free emission from this XeO* state is known to occur at 237 nm. Unfortunately neither photolytic decomposition of XeO_4 nor metastable rare gas atom reaction gave XeO** emission. Perhaps the most valuable result from this study was the development of a reasonably good and safe gas phase source of XeO_4.

We investigated the $Xe(^3P_2)$ and $Ar(^3P_{0,2})$ reactions with XeO_4 as a way of obtaining a strong source of XeO** emission because no good source is available. For these reactions, KrF_2 serves as a good F atom donor and gives high yield of XeF(B,C) upon reaction with $Xe(^3P_2)$. On the other hand, XeF_2 primarily reacts by excitation transfer followed by dissociation in reactions with metastable atoms. Since there are two mechanisms, either excitation transfer followed by dissociation or reactive quenching via the charge transfer intermediate, it was hoped that these reactions would give XeO. Unfortunately these hopes were not realized and we still do not have a strong emission source of the XeO** emission. The only known sources are the reactions of N_2O (and probably O_3) with $Xe(^3P_2)$ aotms. We did observe weak $O_2(B^3\Sigma_u^- - X^3\Sigma_g^+)$ emission from the $Ar(^3P_{0,2})$ reaction with XeO_4. In the following pages the techniques developed to produce XeO_4 and the O_2(B-X) emission spectrum are discussed.

Experimental

The XeO_4 was prepared by the reaction of Ba_2XeO_6 with H_2SO_4 at 0°C. The Ba_2XeO_6 crystals were placed in a side arm of an evacuated 500 ml vessel containing concentrated H_2SO_4. From 0.5 to 15 torr could be generated in the reservoir by allowing Ba_2XeO_6 to drop slowly into the acid. The XeO_4 was

then metered from this reservoir. This reaction of $BaXeO_6$ is a tested method for generating gaseous XeO_4 and was, in fact, the method used to obtain the infrared absorption spectrum of XeO_4. In our experience the method is quite satisfactory.

The experimental flow apparatus used for the $Ar(^3P_2)$ and $Xe(^3P_2)$ reactions has been described previously. The XeO_4 generator was connected directly to the reagent inlet of the rare gas flow; the inlet terminated at the coaxial mixing zone located $\sim$ 5 cm downstream of the hollow cathode discharge. Any emission from the mixing zone was detected with a 0.75 m Jarrell-Ash Czerny Turner scanning monochromator equipped with an E.M.1. 9558QB photomultiplier. Spectra could be stored on magnetic tape by a PDP/8e computor interfaced to the monochromator. Spectra are corrected for spectral response of the detection system.

The vac. U.V. photolysis lamps were driven with a microwave discharge. Experiments were done with a Kr(123.6), Xe(147.0) and CO(160-180nm) resonance lamps. The cell was viewed at right angles to the excitation light by a 1/2 meter Jarrell-Ash monochromator. The XeO_4 was introduced into a rectangular stainless steel reaction cell, which was interfaced to the UV lamps via an LiF window.

The Ba_2XeO_4 was prepared by the method mentioned in the previous quarterly report. This is repeated in more detail below.

The synthesis of XeO_3 from XeF_6 involves the following reaction:

$$XeF_6 + 3\ HOPOF_2 \rightarrow XeO_3 + 3HF + 3OPF_3$$

A 2:1 excess of $HOPOF_2$ was used, providing completeness of reaction and insuring that the XeO_3 did not prematurely dry out. Precautions were taken against XeO_3 detonation. The experimenter wore heavy leather welder's gloves, lab coat and safety glasses in addition to a plastic face shield. Heavy plexiglass shields were placed between the apparatus and the experimenter. The synthesis was performed on a vacuum line utilizing a 150 ml round bottom flask fitted with a teflon stopcock.

A sidearm on the flask would be connected to a 50 ml round bottom flask with a Fischer-Porter stopcock. A Swagelok fitting on the sidearm allowed separation of the flasks. The 50 ml flask could be filled with NaOH or $Ba(OH)_2$ solution. The 150 ml flask was also connected to one side of a Kel-F T-joint on a stainless steel vacuum line. The other side of the T-joint could be connected to a Kel-F tube containing XeF_6. A Kel-F U-trap was also connected to the vacuum line in a swing-away position, in order to have the open end ready to replace the XeF_6 tube.

For the preparation of the XeO_3 the entire system was evacuated, and the 150 ml reaction bulb was dryed with a heat gun. Approximately 1 mmole of XeF_6 was condensed into the bulb at 77°K, making a spot the size of a half-dollar or less. This was done to insure no XeO_3 forms on the walls upon introduction of the $HOPOF_2$. The XeF_6 tube was then removed and replaced with the Kel-F U-trap to vacuum. The level of liquid N_2 was then raised around the reaction bulb, and $\sim$ 0.8 cc ($\sim$ 6 mmoles) of $HOPOF_2$ was condensed in a ring above the XeF_6. The $HOPOF_2$ was metered from a graduated cylinder fitted with a teflon stopcock. The reaction bulb was then allowed to warm up. As the sample warms, $HOPOF_2$ liquid runs down into the XeF_6, creating a yellow-green liquid. At $\sim$ 0°C, vigorous bubbling became evident, followed by rapid precipitation of white XeO_3 crystals. The Kel-F U-trap was then cooled to 77°K, and a 0°C bath placed under the reaction bulb, and the excess HF, $HOPOF_2$ and OPF_3 could be slowly pumped off until the XeO_3 is nearly dry. To avoid detonation, the XeO_3 should not be completely dried. The bulb was once again isolated and the 0°C bath replaced by liquid N_2. Approximately 20 ml of saturated $Ba(OH)_2$ solution was then placed in the smaller (50 ml) flask, and cooled to 0°C to avoid thermal shock. The sidearm stopcock was then opened slowly and the vacuum allowed to draw the hydroxide solution into the reaction bulb, stopping before the last of the solution was introduced. The liquid N_2 bath was then removed and the bulb allowed to warm to room temperature. The room temperature solution had an irridescent yellow color indicative of the perxenate.

The solution was poured into an Erlynmeyer flask and left to disproportionate for an indefinite period. The half-times of the disproportionation reaction vary, depending on base concentration and concentration of the Xe species. The exact route of the disproportionation reaction is not known, but one suggested possibility is

$$XeO_3 + 6OH^- \rightarrow 3HXeO_6^{-3} + Xe + H_2O.$$

The barium penzonate product was then filtered, washed with H_2O at 0°C and heated to 100°C in order to drive off H_2O.

High purity was not necessary to eventually generate XeO_4, since it is the only gaseous product. Therefore percent yield was not determined. Potency of the perxenate mixture was determined by iodometric titration. Excess

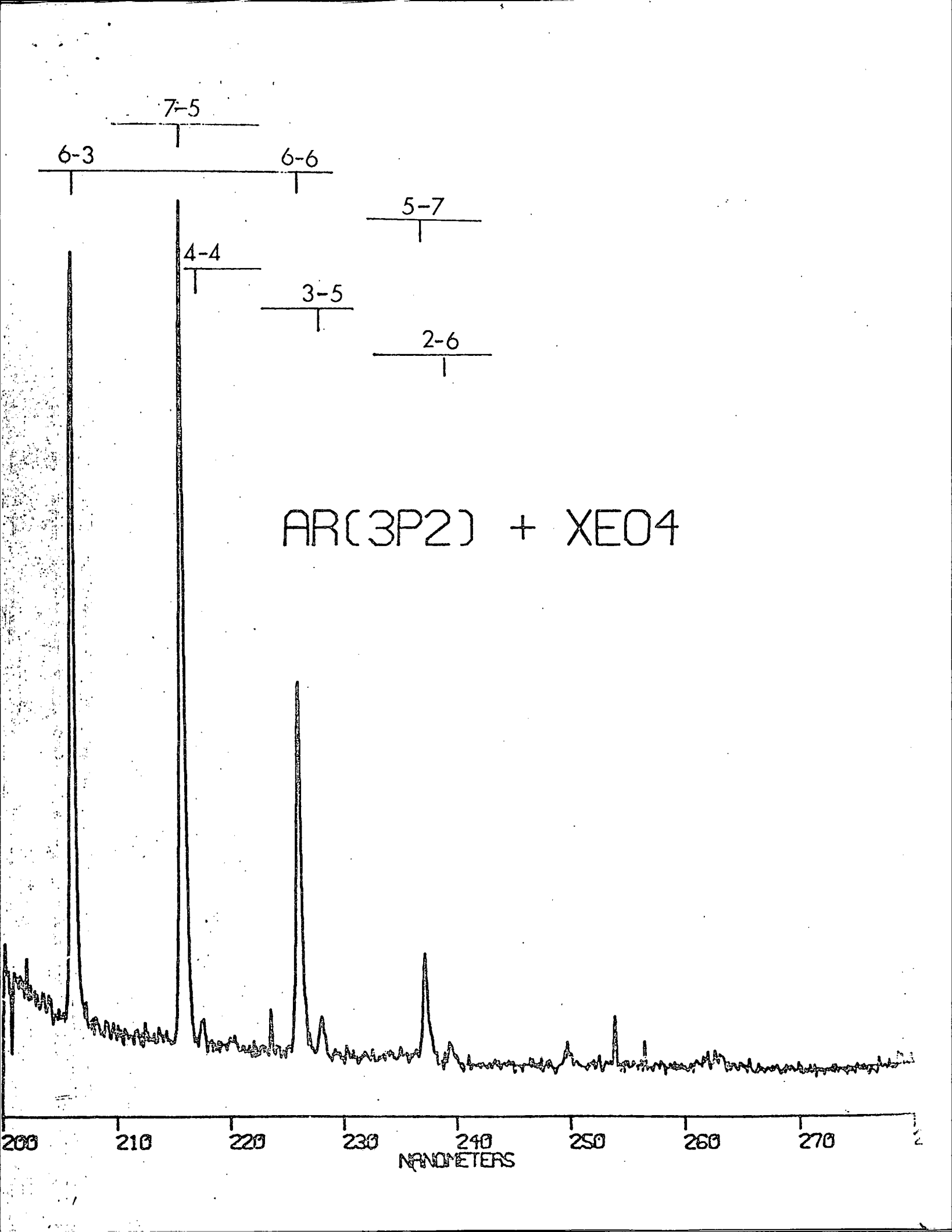
acidified KI solution (0.45 M) was added to a known weight of the mixture. The I_2 was back-titrated with 0.42 M $Na_2S_2O_3$ until the red color disappeared. Calculations show an average of 88.4 g-% XeO_6^{-4} per gram of solid compound.

Results

Irradiation of XeO_4 (0.5 to 10 torr) by Kr(123.6 nm), Xe(147.0 nm) and CO(160-180 nm) resonance radiation yielded no observable emission from 200 nm to 500 nm. There was, however, a pressure rise upon photolysis. The pressure increase presumably results from $XeO_4 \rightarrow Xe + 2O_2$. The pressure rise was slow, never more than 60% in one hour, and varied for irradiation with each lamp. Since the pressure rise was less than 10% per hour without irradiation, the major part of the pressure increase must be from photo-decomposition of XeO_4. Gaseous XeO_4 is thermally unstable at room temperature and slowly reverts back to Xe and O_2. The variation of the rate of decomposition from irradiation with different lamps may be due either to the variation in lamp emission intensity or to a variation in the absorption coefficients of XeO_4 at various wavelengths.

In the reaction of XeO_4 with $Ar(^3P_2)$ and $Xe(^3P_2)$, the spectral range from 200 - 500 nm was investigated; however, only with $Ar(^3P_2)$ was any emission observed. Weak $O_2(b^3\Sigma_u^- - X^3\Sigma_g^-)$ emission and weak Xe* emission was found. The Xe* emission most likely was the result of reaction of Ar* with trace amounts of Xe in the XeO_4. The vibrational distribution of $O_2(B^3\Sigma_u^-)$ was rather high, the most intense emission was from the v = 6,7 levels. The v = 5,4,3 and 2 were observed, but were weak compared to the v = 6,7, comprising (see Figure 1). Careful searches were made for the $O(^1S - ^1D)$ transition at 557.7 nm and the $O(^1D - ^3P)$ transition at 636.4 nm, but, these could not be observed. Searches also were made for $O_2(A,C,c)$ state emission. These three states are long lived states and if the concentration was as low as $O_2(B)$ they may not have been observable. However, large concentrations of $O_2(A,C,c)$ obviously were not generated by reaction of $Xe(^3P_2)$ or $Ar(^3P_2)$ with XeO_4.

We now have investigated XeO_4, N_2O, OsO_4, CrO_2Cl_2 and O_3 as oxygen atom donors for formation of the coulombic ArO*, KrO* and XeO* states. None of these molecules are good oxygen atom donors; only N_2O and O_3 give the rare gas oxides and the yields are extremely low, $\sim$1%. The inescapable conclusion seems to be that efficient formation of XeO* by reactive quenching is not possible. Other excitation schemes, such as ion recombination may be more promising.


7-5
6-3
6-6
5-7
4-4
3-5
2-6
AR(3P2) + XE04
200
210
220
230
240
250
260
270
NANOMETERS

Cumulative Listing of Pulbished Papers

1. J. E. Velazco, J. H. Kolts and D. W. Setser, "Quenching rate constants for metastable argon, krypton, and xenon atoms by fluorine containing molecules and branching ratios for XeF and KrF formation," J. Chem. Phys., 65, 3468 (1976).

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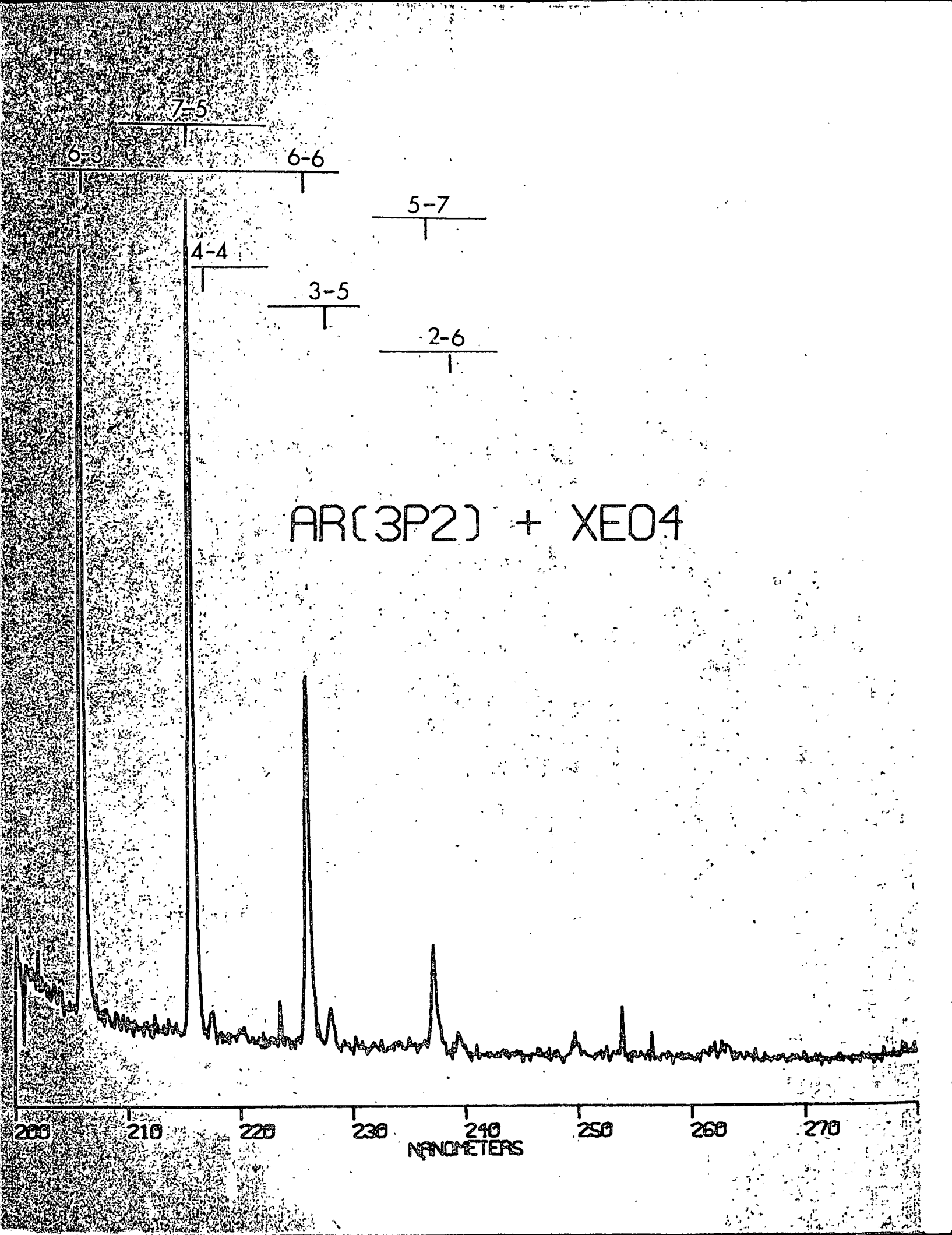
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